

1. $M_i \in M_n(\mathbb{C})$. $M_i = A_i + \sqrt{-1} B_i$, $i=1, 2$,
with $A_i, B_i \in M_n(\mathbb{R})$

$$T(M_i) := \begin{pmatrix} A_i & -B_i \\ B_i & A_i \end{pmatrix}, \text{ then}$$

$$T(M_1 + M_2) = T(M_1) + T(M_2)$$

$$T(M_i^H) = T(M_i)^T.$$

Remark: M normal $\Leftrightarrow T(M)$ normal.

M Hermite $\Leftrightarrow$ real symmetric.

M unitary $\Leftrightarrow$ orthogonal.

Proof: Calculate directly.

$$\begin{aligned} (1) \quad T(M_1 + M_2) &= T\left((A_1 + A_2 - B_1 B_2) + \sqrt{-1} (A_1 B_2 + B_1 A_2)\right) \\ &= \begin{pmatrix} A_1 + A_2 - B_1 B_2 & -A_1 B_2 - B_1 A_2 \\ A_1 B_2 + B_1 A_2 & A_1 + A_2 - B_1 B_2 \end{pmatrix} \\ T(M_1) + T(M_2) &= \begin{pmatrix} A_1 & -B_1 \\ B_1 & A_1 \end{pmatrix} + \begin{pmatrix} A_2 & -B_2 \\ B_2 & A_2 \end{pmatrix} \end{aligned}$$

$$\begin{aligned} (2) \quad T(M_i^H) &= T(A_i^T - i B_i^T) = \begin{pmatrix} A_i^T & B_i^T \\ -B_i^T & A_i^T \end{pmatrix} \\ &= T(M_i)^T. \end{aligned}$$

□

How to understand :

$$\mathbb{C} \rightsquigarrow M_2(\mathbb{R}).$$

View $z \in \mathbb{C}$ as a linear map :

$$T_z : \mathbb{C} \rightarrow \mathbb{C} \quad / \quad \mathbb{R}^2 \rightarrow \mathbb{R}^2 \\ z' \mapsto z \cdot z'.$$

$\mathbb{C} = \mathbb{R} \cdot 1 \oplus \mathbb{R} \cdot \sqrt{-1}$, "forget" T_z $\mathbb{C}$ -linear.
Under basis $\{1, \sqrt{-1}\}$, T_z is represented
by matrix $\begin{pmatrix} a & b \\ -b & a \end{pmatrix}$.

$$\begin{pmatrix} z \cdot 1 = a \cdot 1 + b \cdot \sqrt{-1} \\ z \cdot i = -b \cdot 1 + a \cdot \sqrt{-1} \end{pmatrix}$$

$\rightarrow$ For $\mathbb{C}^n$, it has $\mathbb{C}$ -basis
 $e_1, \dots, e_n$; $\mathbb{C}^n \cong \mathbb{R}^{2n}$ with basis
 $e_1, \dots, e_n, \sqrt{-1}e_1, \dots, \sqrt{-1}e_n$.

Conclusion: The bijection above is just.
to forget the complex structure.

2. Generalize. $z = r \cdot e^{i\theta}$ to $A \in M_n(\mathbb{C})$.

idea: view z as a (linear transform)
 $T_z := \mathbb{C} \rightarrow \mathbb{C}$
 $z' \mapsto z \cdot z'$
 $A: \mathbb{C}^n \rightarrow \mathbb{C}^n$
 $\nearrow ?$
 then $T_z = \text{Scalar} \circ \text{"Rotation"}$.

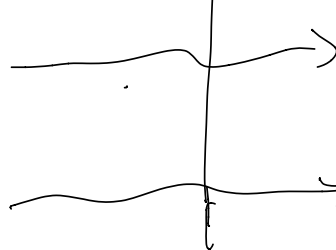
$\mathbb{C}$ -case:

how to get r, θ from z ?

$$r = \sqrt{z \cdot \bar{z}}$$

$$e^{i\theta} = r^{-1} z$$

$\mathbb{C}^n$ -case:



$$A \cdot A^H$$

$$= (A \cdot A^H)^{-1/2} \cdot A$$

(1). eigenvalue of $A \cdot A^H > 0$.

Assume x is an eigenvector of $A \cdot A^H$

λ is its eigenvalue.

$$x^H \cdot A \cdot A^H \cdot x = y^H y \geq 0.$$

" $\equiv$ " iff $y = 0$ iff $x = 0$ (A invertible)

$$x^H \cdot A \cdot A^H \cdot x = \lambda x^H \cdot x \Rightarrow \lambda = \frac{y^H y}{x^H x} > 0.$$

□

(2). $A \cdot A^H$ Hermitian $\Rightarrow A \cdot A^H = U(\lambda_1, \dots, \lambda_n) \cdot U^H$
 with $\lambda_i > 0$ by (1).

Set $R = U(\sqrt{\lambda_1}, \dots, \sqrt{\lambda_n}) \cdot U^H$,
 then $R^2 = A \cdot A^H$. $\leadsto$ (order of λ_i ,
 choice of U
 are not canonical)

(Uniqueness) Notice that

$$f(x) := \sum_{i=1}^t \lambda_i^{\frac{1}{2}} \prod_{\substack{k=1 \\ k \neq i}}^t \frac{\lambda_i - \lambda_k}{\lambda_i - \lambda_k}$$

(here $\lambda_1, \dots, \lambda_t$ are distinct eigenvalues
 of $A \cdot A^H$.)

s.t. $f(\lambda_i) = \sqrt{\lambda_i} \quad \forall i.$

$\Rightarrow f(A \cdot A^H) = R$, R is determined by $A \cdot A^H$. $\square$

(3). $V := R^{-1} A$. Only need to show
 V is unitary.

$$V^H = (R^{-1} A)^H = A^H \cdot R^{-1}$$

$$V \cdot V^H = R^{-1} A \cdot A^H \cdot R^{-1} = I_n. \quad \square$$

Why does $f(AA^H) = R$ imply uniqueness?

Suppose $R'^2 = A \cdot A^H = R^2$ with
eigenvalue > 0 .

$$\exists U, V \text{ s.t.}$$

$$UR'U^H = \begin{pmatrix} \beta_1 & & \\ & \ddots & \\ & & \beta_n \end{pmatrix} \quad \beta_1 \sim \beta_t \text{ distinct}$$

$$VRV^H = \begin{pmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{pmatrix} \quad \lambda_1 \sim \lambda_s \text{ distinct}$$

$$\text{Define } f_{R'}(x) = \sum_{i=1}^t \left(\beta_i \prod_{\substack{k=1 \\ k \neq i}}^t \frac{x - \beta_k^2}{\beta_i^2 - \beta_k^2} \right)$$

$$f_R(x) = \sum_{i=1}^s \left(\lambda_i \prod_{\substack{k=1 \\ k \neq i}}^s \frac{x - \lambda_k^2}{\lambda_i^2 - \lambda_k^2} \right)$$

$$\text{then } \underline{f_{R'}(A \cdot A^H) = R'} \quad ?$$

$$\underline{f_R(A \cdot A^H) = R}$$

$\therefore \{ \beta_1^2, \dots, \beta_n^2 \} = \{ \lambda_1^2, \dots, \lambda_n^2 \}$
is the set of eigenvalue of AA^H

$$\therefore f_{R'} = f_R \Rightarrow R = R' \quad \square.$$

$$? : U R' U^H = \begin{pmatrix} \beta_1 & & \\ & \ddots & \\ & & \beta_n \end{pmatrix}$$

$$U A A^H U^H = (U R' U^H)^2 = \begin{pmatrix} \beta_1^2 & & \\ & \ddots & \\ & & \beta_n^2 \end{pmatrix}$$

$$\text{tr}_{R'}(A A^H) = \text{tr}_{R'}(U^H \begin{pmatrix} \beta_1^2 & & \\ & \ddots & \\ & & \beta_n^2 \end{pmatrix} U)$$

$$= U^H \text{tr}_{R'} \left(\begin{pmatrix} \beta_1^2 & & \\ & \ddots & \\ & & \beta_n^2 \end{pmatrix} \right) U$$

$$= U^H \begin{pmatrix} \text{tr}_{R'}(\beta_1^2) & & \\ & \ddots & \\ & & \text{tr}_{R'}(\beta_n^2) \end{pmatrix} U$$

$$= U^H \begin{pmatrix} \beta_1 & & \\ & \ddots & \\ & & \beta_n \end{pmatrix} U$$

$$= R' \quad \square.$$

(4). Replace U by V ,

Since V unitary, $\exists W$ s.t.

$$W^H V W = \begin{pmatrix} \alpha_1 & & \\ & \ddots & \\ & & \alpha_n \end{pmatrix}$$

with $|\alpha_i| = 1 \quad \forall i$.

$$\text{Set } H = W \cdot \begin{pmatrix} \theta_1 & & \\ & \ddots & \\ & & \theta_n \end{pmatrix} \cdot W^H,$$

where θ_j satisfies $e^{i\theta_j} = \alpha_j \quad \forall j$.

$$\text{then } e^{iH} = W \begin{pmatrix} e^{i\theta_1} & & \\ & \ddots & \\ & & e^{i\theta_n} \end{pmatrix} W^H$$

$$= V.$$

□

Conclusion: A invertible $\in M_n(\mathbb{C})$.

$\exists R, H$ s.t.

$$(1) \quad A = R \cdot e^{iH}.$$

(2) R Hermite, with positive eigenvalues.

(3) H Hermite (R unique, but H has many choice).

* Series of matrix:

$$e^A, \log A, \dots ?$$

$$e^x = 1 + x + \frac{x^2}{2!} + \dots$$

$$e^A := 1 + A + \frac{A^2}{2!} + \dots$$

($\log A$ similarly.)

(1) We need that the coefficients of $e^A, \log A$ are convergent. $\begin{pmatrix} (8.5) \text{ for} \\ \log A \end{pmatrix}$

(2) In exercise, $U(\lambda_1, \dots, \lambda_n) \cdot U^H$.

$$e^{U(\lambda_1, \dots, \lambda_n) \cdot U^H} = \sum_{n=0}^{\infty} \frac{(U(\lambda_1, \dots, \lambda_n) U^H)^n}{n!}$$

$$= \sum_{n=0}^{\infty} \frac{U \cdot \begin{pmatrix} \lambda_1^n & & \\ & \ddots & \\ & & \lambda_n^n \end{pmatrix} \cdot U^H}{n!}$$

$$= U \begin{pmatrix} e^{\lambda_1} & & \\ & \ddots & \\ & & e^{\lambda_n} \end{pmatrix} \cdot U^H$$

coincide with previous definition.

$$3. \quad u, v \in \mathbb{C}^n, \quad \langle u, v \rangle = \frac{1}{4} \left(\|u+v\|^2 - \|u-v\|^2 \right. \\ \left. + i \|u-iv\|^2 - i \|u+iv\|^2 \right),$$

$$\text{Proof:} \quad \|u+v\|^2 - \|u-v\|^2 \\ = \left(\langle u, v \rangle + \langle v, u \rangle + \|u\|^2 + \|v\|^2 \right) \\ - \left(\|u\|^2 + \|v\|^2 - \langle u, v \rangle - \langle v, u \rangle \right) \\ = 2(\langle u, v \rangle + \langle v, u \rangle) \quad (1)$$

$$i \|u-iv\|^2 - i \|u+iv\|^2 \\ = \left(i \|u\|^2 + i \|v\|^2 + \langle u, v \rangle - \langle v, u \rangle \right) \\ - \left(i \|u\|^2 + i \|v\|^2 + \langle v, u \rangle - \langle u, v \rangle \right) \\ = 2(\langle u, v \rangle - \langle v, u \rangle) \quad (2)$$

$$\frac{(1) + (2)}{4}.$$

done!

□

$$4. \quad V = \{a_0 + a_1 x + a_2 x^2 \mid a_i \in \mathbb{C}\}.$$

$$\langle f, g \rangle := \int_{-1}^1 \overline{f} \cdot g \, dx.$$

(1). $\langle -, - \rangle$ is an inner product.

(2) GS and orthogonal basis.

(3) $u(x) \in V$ s.t.

$$\int_{-1}^1 p(x) (\cos \pi x - u(x)) \, dx = 0.$$

$$\text{Proof: (1) } \langle f, f \rangle = 0 \Leftrightarrow \int_{-1}^1 |f|^2 \, dx = 0.$$

$$\Leftrightarrow |f|^2 = 0 \quad \text{a.e.}$$

$$\Leftrightarrow f = 0.$$

$$(2) \quad \alpha_1 = 1 \quad \alpha_2 = x \quad \alpha_3 = x^2.$$

$$\beta_1 = \alpha_1, \quad \|\beta_1\| = \int_{-1}^1 1^2 \, dx = 2.$$

$$e_1 = \frac{1}{\sqrt{2}} \cdot 1.$$

$$\begin{aligned}
 \beta_2 &= \alpha_2 - \langle e_1, \alpha_2 \rangle e_1 \\
 &= x - \int_{-1}^1 \frac{1}{\sqrt{2}} x \, dx \cdot \frac{1}{\sqrt{2}} \\
 &= x.
 \end{aligned}$$

$$||\beta_2|| = \int_{-1}^1 x^2 \, dx = \frac{2}{3}$$

$$\Rightarrow e_2 = \sqrt{\frac{3}{2}} x.$$

$$\begin{aligned}
 \beta_3 &= \alpha_3 - \langle e_1, \alpha_3 \rangle e_1 - \langle e_2, \alpha_3 \rangle e_2 \\
 &= x^2 - \int_{-1}^1 \frac{x^2}{\sqrt{2}} \, dx \cdot \sqrt{\frac{1}{2}} \\
 &= x^2 - \frac{1}{3}.
 \end{aligned}$$

$$||\beta_3|| = \sqrt{\frac{8}{45}} \Rightarrow e_3 = \sqrt{\frac{45}{8}} \left(x^2 - \frac{1}{3}\right).$$

$$(3) \quad \langle e_1, \cos \pi x \rangle$$

$$= \int_{-1}^1 \sqrt{\frac{1}{2}} \cos \pi x \, dx = 0$$

$$\langle e_2, \cos \pi x \rangle = 0$$

$$\langle e_3, i\pi X \rangle$$

$$= \int_{-1}^1 \sqrt{\frac{45}{8}} x^2 i\pi x dx$$

$$= - \frac{\sqrt{90}}{\pi^2}.$$

$$\Rightarrow U(X) = -\frac{45}{2\pi^2} \left(X^2 - \frac{1}{3} \right). \quad \square$$

5. (1) $\mathbb{C}^4 \rightarrow \mathbb{C}^4$ normal, not Hermitian.

e.g. $T: \mathbb{C}^4 \rightarrow \mathbb{C}^4 \quad x \mapsto i \cdot x.$

(2) $T \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 2 \\ 2 \\ 2 \\ 2 \end{pmatrix}$, T normal.

$$TV = 0 \Rightarrow z_1 + z_2 + z_3 = 0$$

Proof: $T \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix} = 2 \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix}.$

$$T \begin{pmatrix} z_1 \\ z_2 \\ z_3 \\ z_4 \end{pmatrix} = 0 \cdot \begin{pmatrix} z_1 \\ z_2 \\ z_3 \end{pmatrix}$$

$$z \neq 0 \Rightarrow (1 \ 1 \ 1), (z_1, z_2, z_3) \\ \text{orthogonal}$$

$$\Rightarrow z_1 + z_2 + z_3 = 0.$$

$$6. \quad \langle f, g \rangle := \int_{-\pi}^{\pi} f \cdot g \, dx$$

$$(1) \quad D: V \rightarrow V \quad D(f) = f'$$

$$\text{then } D^* = -D.$$

$$(2) \quad T(f) = f'' \quad \text{then } T^* = T.$$

Proof: (check for basis).

$$(1) \quad \forall f, g.$$

$$\langle Df, g \rangle = \langle f, D^*g \rangle.$$

$$\langle Df, g \rangle = \int_{-\pi}^{\pi} f' g \, dx$$

$$= \int_{-\pi}^{\pi} (fg)' - fg' \, dx$$

$$= \underbrace{fg \Big|_{-\pi}^{\pi}} - \int_{-\pi}^{\pi} f(x) g'(x) \, dx$$

For any f, g in basis of V ,

$$fg \Big|_{-\pi}^{\pi} = 0 \quad \text{always holds.}$$

$$\Rightarrow \langle Df, g \rangle = \langle f, -g' \rangle$$

$$\Rightarrow \langle f, -g' \rangle = \langle f, D^*g \rangle$$

$$\therefore D^*g = -g' = -Dg$$

$$(2) \quad T = D^2 \quad T^* = (D^*)^2$$

$$= (-D)^2$$

$$= D^2 = T \quad \square$$

$$7. \quad (1). \quad v^H A v = 0 \quad \forall v \in \mathbb{C}^n \\ \Rightarrow A = 0.$$

$$(2). \quad v^T A v = 0 \quad \forall v \in \mathbb{R}^n \\ \nRightarrow A = 0.$$

Proof: (1). ($u^H A v = 0 \Rightarrow A = 0$)
Similar as 3.

$$\begin{aligned} & \langle u+v, A(u+v) \rangle - \langle u-v, A(u-v) \rangle \\ &= (\langle u, Au \rangle + \langle v, Av \rangle + \langle v, Au \rangle + \langle u, Av \rangle) \\ & \quad - (\langle u, Au \rangle + \langle v, Av \rangle - \langle v, Au \rangle - \langle u, Av \rangle) \\ &= 2(\langle v, Au \rangle + \langle u, Av \rangle). \quad (1) \end{aligned}$$

$$\begin{aligned} & i\langle u+iv, A(u+iv) \rangle - i\langle u-iv, A(u-iv) \rangle \\ &= 2(\langle v, Au \rangle - \langle u, Av \rangle). \quad (2) \end{aligned}$$

$$\text{Let } y_1 = u+v \quad y_2 = u-v$$

$$y_3 = u+iv \quad y_4 = u-iv.$$

$$\begin{aligned} u^H A v &= y_1^H A y_1 - y_2^H A y_2 \\ &\quad + i \cdot y_4^H A y_4 - i y_3^H A y_3. \end{aligned}$$

$$= 0 \quad \forall u, v.$$

$$\Rightarrow A = 0.$$

□.

$$(2). \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \in M_2(\mathbb{R}).$$

$$\forall v \in \mathbb{R}^2. \quad v^T \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \cdot v = 0.$$

Remark: $\langle -, - \rangle$ is a $\mathbb{C}$ -linear form of V .

$$\text{S.t.} \quad \langle x, x \rangle = 0 \quad \forall x.$$

$$\text{Define } *: V \rightarrow \text{Hom}_{\mathbb{C}}(V, \mathbb{C}).$$

$$x \mapsto (*.x : V \rightarrow \mathbb{C})$$

$$y \mapsto \langle \bar{y}, x \rangle$$

then $*$ is zero, i.e. $\langle x, y \rangle = 0$
 $\forall x, y.$

8. $\{e_1, e_2, e_3\}$. $\{v_1, v_2, v_3\}$.

$$\|e_j - v_j\| < \frac{1}{\sqrt{3}}. \quad \text{then}$$

$\{v_1, v_2, v_3\}$ form a basis

Proof: Suppose $\exists (a_1, a_2, a_3) \neq 0$.

$$\text{s.t.} \quad a_1 v_1 + a_2 v_2 + a_3 v_3 = 0$$

$$|a_1|^2 + |a_2|^2 + |a_3|^2$$

$$= \|a_1 e_1 + a_2 e_2 + a_3 e_3\|.$$

$$= \|a_1(e_1 - v_1) + a_2(e_2 - v_2) + a_3(e_3 - v_3)\|$$

$$= \left| \sum_{i,j=1}^3 \overline{a_i} a_j \langle e_i - v_i, e_j - v_j \rangle \right|$$

$$\leq \sum_{i,j=1}^3 |a_i| |a_j| |\langle e_i - v_i, e_j - v_j \rangle|$$

$$\leq \sum_{i,j=1}^3 |a_i| |a_j| \|e_i - v_i\| \|e_j - v_j\|.$$

$$< \frac{1}{3} \cdot \sum_{i,j=1}^3 |a_i| \cdot |a_j|.$$

$$\leq |a_1|^2 + |a_2|^2 + |a_3|^2.$$

$$\left(\text{using } 2|a_i| |a_j| \leq |a_i|^2 + |a_j|^2, \quad i \neq j \right).$$

Contradict!

□.

How to understand it?

$$\text{Consider } A \begin{pmatrix} e_1 \\ e_2 \\ e_3 \end{pmatrix} = \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix},$$

$$\text{when } v_i = e_i. \quad A = I_3.$$

$$\text{but } v_i \text{ is close to } e_i.$$

$$\Rightarrow A \text{ is "close to" } I_3.$$

$$\Rightarrow \det A \text{ is close to } 1 \Rightarrow A \text{ invertible.}$$

$$8. \quad \sum |\lambda_i|^2 \leq \sum |a_{ij}|^2.$$

"=" iff A normal.

Proof: $\exists U$ unitary s.t.

$$U^H A U = \begin{pmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{pmatrix} := B.$$

Set $B^H B = (C_{ij})$.

$$\text{then } C_{ii} = \sum_{k=1}^{i-1} |b_{ki}|^2 + |\lambda_i|^2.$$

Set $A^H A = (d_{ij})$.

$$\text{then } d_{ii} = \sum_{k=1}^n |a_{ki}|^2$$

$$\begin{aligned} \text{tr}(A^H A) &= \sum_{i=1}^n \sum_{k=1}^n |a_{ki}|^2 \\ &= \sum_{i,j} |a_{ij}|^2 \end{aligned}$$

$$\text{tr}(A^H A) = \text{tr}(U B^H U^H U B U^H)$$

$$= \text{tr}(B^H B)$$

$$= \sum_{i=1}^n |\lambda_i|^2 + \sum_{k < i} |b_{ki}|^2$$

$$\Rightarrow \sum |a_{ij}|^2 = \sum |\lambda_i|^2 + \sum |b_{ki}|^2$$

$$\text{i.e.} \quad \sum |\lambda_i|^2 \leq \sum |a_{ij}|^2$$

$$\text{"="} \Leftrightarrow b_{ki} = 0$$

$$\Leftrightarrow U^H A U \text{ diagonal}$$

$$\Leftrightarrow A \text{ is normal} \quad \square$$

Hw : $T: \mathbb{C}^m \rightarrow \mathbb{C}^n$.

$$T \begin{pmatrix} e_1 \\ \vdots \\ e_n \end{pmatrix} = \begin{pmatrix} 1 & & \\ & 2 & \\ & & \ddots \\ & & & n \end{pmatrix} \begin{pmatrix} e_1 \\ \vdots \\ e_n \end{pmatrix}.$$

determine all T -invariant subspace.

$V = \text{Span} \{ e_{i_1}, \dots, e_{i_k} \} \quad 0 \leq k \leq n.$ is T -inv.

Why no others?

Suppose V' is a T -inv. subspace.

take $v = a_{i_1} e_{i_1} + \dots + a_{i_k} e_{i_k} \in V'.$

with $a_{i_j} \neq 0$ and $T e_{i_r} = \lambda_r e_{i_r}$

Then $Tv, T^2v, \dots, T^{k-1}v \in V'.$

$$T^j v = \lambda_1^j a_{i_1} e_{i_1} + \lambda_2^j a_{i_2} e_{i_2} + \dots + \lambda_k^j a_{i_k} e_{i_k}$$

i.e.

$$(v, Tv, \dots, T^{k-1}v)^t = \begin{pmatrix} a_{i_1} & \dots & a_{i_k} \\ \lambda_1 a_{i_1} & \dots & \lambda_k a_{i_k} \\ \vdots & & \vdots \\ \lambda_1^{k-1} a_{i_1} & \dots & \lambda_k^{k-1} a_{i_k} \end{pmatrix} \begin{pmatrix} e_{i_1} \\ \vdots \\ e_{i_k} \end{pmatrix}$$

$M \triangleq$

$$\det M = a_{11} a_{22} \cdots a_{kk} \cdot \prod_{i < j} (\lambda_i - \lambda_j) \neq 0.$$

$$\Rightarrow \text{Span} \{v, Tv, \dots, T^{k-1}v\}$$

$$= \text{Span} \{e_{i_1}, \dots, e_{i_k}\}.$$

$$\Rightarrow \text{Span} \{e_{i_1}, \dots, e_{i_k}\} \subseteq V'$$

$$\text{If } V' = \text{Span} \{e_{i_1}, \dots, e_{i_k}\}, \text{ done!}$$

Otherwise, take $v' \in V'$,

$$v' \notin \text{Span} \{e_{i_1}, \dots, e_{i_k}\}.$$

we can add some e_{i_1} until

$$\text{Span} \{e_{i_1}, \dots, e_{i_k'}\} = V'. \quad \square.$$

$$\longrightarrow |\{T\text{-inv spaces}\}| = 2^n.$$

$$\longrightarrow |\{m\text{-dim } T\text{-inv spaces}\}| = \binom{n}{m}.$$

Q: What if $\lambda_i = \lambda_j$?

Prop: $T: V \rightarrow V$ is diagonalizable.

$$V = V_{\lambda_1} \oplus \dots \oplus V_{\lambda_k} \text{ with } \lambda_i \neq \lambda_j.$$

then any T -inv. subspace W has

$$\text{decomposition } W = \bigoplus_{i=1}^k W_{\lambda_i}$$

$$\text{where } W_{\lambda_i} = W \cap V_{\lambda_i}.$$

Proof: The proof of hw still holds.

For any $w \in W$, $w = a_1 w_1 + \dots + a_s w_s$

$$\text{with } T w_i = \lambda_i w_i, \quad \lambda_i \neq \lambda_j$$

$$(w, T w, \dots, T^{s-1} w) = M \begin{pmatrix} w_1 \\ \vdots \\ w_s \end{pmatrix}$$

$$\Rightarrow w_1, w_2, \dots, w_s \in W.$$

$$\Rightarrow W \subseteq \bigoplus_{i=1}^k (W \cap V_{\lambda_i}). \quad \text{"}\supseteq\text{" trivial}$$

□.

So we only need to deal with
subspaces of V_λ .

$$T: V \rightarrow V \quad \dim V = n.$$
$$v \mapsto \lambda \cdot v.$$

$$\Rightarrow \{T\text{-inv. subspace}\} \Leftrightarrow \{\text{subspace of } \mathbb{C}^n\}$$

Now can we ask 'how many?'

baby case: $T: \mathbb{C}^2 \rightarrow \mathbb{C}^2$

$$v \mapsto \lambda \cdot v.$$

- 0-dim $\{0\}$
- 2-dim $\mathbb{C}^2$
- 1-dim $\text{span}\{a_1 e_1 + a_2 e_2\} := V(a_1, a_2).$

Since $\text{span}\{v\} = \text{span}\{cv\}$ for $c \neq 0$.

$$\text{so } V(a_1, a_2) = V(ca_1, ca_2)$$

1-dim subspace of $\mathbb{C}^2$
 can be identified with

$$\mathbb{C}^2 - \{(0,0)\} \nearrow$$

$$(a_1, a_2) \sim (ca_1, ca_2)$$

for $c \neq 0 \in \mathbb{C}$

why $(a_1, a_2) \neq (0,0)$

$// \Delta$

Denote by $\mathbb{P}^1(\mathbb{C})$.

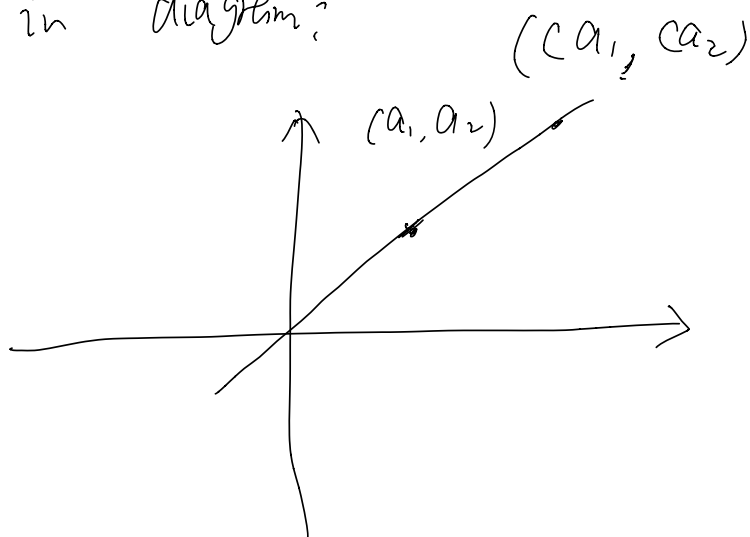
Similarly, $\mathbb{P}^1(\mathbb{R}) := \mathbb{R}^2 - \{(0,0)\} \nearrow$

$$(a_1, a_2)$$

$$\sim (ca_1, ca_2)$$

$0 \neq c \in \mathbb{R}$

$\mathbb{P}^1(\mathbb{R})$ in diagram?



$$\mathbb{P}^1(\mathbb{R}) \xleftrightarrow{1-1} \{ \text{line in } \mathbb{R}^2 \}$$

Generally: $\mathbb{P}^n(\mathbb{C}) := \mathbb{C}^{n+1} - \{0\} / \sim$
 $v \sim cv$
 $0 \neq v \in \mathbb{C}^{n+1}$
 $c \neq 0 \in \mathbb{C}$

$$\mathbb{P}^n(\mathbb{C}) \xleftrightarrow{1-1} \{ 1\text{-dim subspace of } \mathbb{C}^{n+1} \}$$

$$\mathbb{P}^n(\mathbb{R}) \xleftrightarrow{1-1} \{ 1\text{-dim subspace of } \mathbb{R}^{n+1} \}$$

Question: $T: \mathbb{C}^6 \rightarrow \mathbb{C}^6$

$$T \begin{pmatrix} e_1 \\ \vdots \\ e_6 \end{pmatrix} = \begin{pmatrix} 1 & & & & & \\ & 2 & & & & \\ & & 2 & & & \\ & & & 3 & & \\ & & & & 3 & \\ & & & & & 3 \end{pmatrix} \begin{pmatrix} e_1 \\ \vdots \\ e_6 \end{pmatrix}$$

$$\{ 3\text{-dim } T\text{-inv. subspace} \}$$

$$\xleftrightarrow{1-1} ?$$

λ	1	2	3
$\dim V_\lambda$	1	2	3

$$W \subseteq V$$

$$W = W_1 \oplus W_2 \oplus W_3.$$

$\dim$	W_1	W_2	W_3
①	0	0	3
②	0	1	2
③	1	0	2
④	0	2	1
⑤	1	1	1
⑥	1	2	0

$$\textcircled{1} \quad \{0\} \oplus \{0\} \oplus \mathbb{C}^3 \quad \text{only } 1$$

$$\textcircled{6} \quad \mathbb{C} \oplus \mathbb{C}^2 \oplus \{0\} \quad \text{only } 1.$$

$$\textcircled{5}. \quad \underline{\mathbb{C}} \oplus \text{line in } \mathbb{C}^2 \oplus \text{line in } \mathbb{C}^3$$

$\Downarrow$
 $\mathbb{P}^1$

$\Downarrow$
 $\mathbb{P}^2.$

$$\mathbb{P}^1 \times \mathbb{P}^2.$$

$$((a_1, a_2), (b_1, b_2, b_3))$$

$$\Downarrow.$$

$$\text{span} \{ e_1, a_1 e_2 + a_2 e_3, b_1 e_4 + b_2 e_5 + b_3 e_6 \}$$

$$\textcircled{4} \quad \mathbb{P}^2 \ni (b_1, b_2, b_3)$$

$$\Downarrow$$

$$\text{span} \{ e_2, e_3, b_1 e_4 + b_2 e_5 + b_3 e_6 \}$$

$$\textcircled{2}. \quad \mathbb{P}^1 \times \star$$

$$\textcircled{3} \quad \star.$$

$$\star : \{ \mathbb{C}^3 \supseteq W \mid \dim W = 2 \}$$

$$\updownarrow | - |.$$

$$\{ \mathbb{C}^3 \supseteq W^\perp \mid \dim W^\perp = 1 \}$$

$$\text{so } \star = \mathbb{P}^2.$$

In a conclusion, we have:

$$\{ 3\text{-dim } T\text{-inv subspace of } \mathbb{C}^6 \}$$

$$\updownarrow | - |.$$

$$\underbrace{\{pt\}}_{\textcircled{1}} \cup \underbrace{\mathbb{P}^1(\mathbb{C}) \times \mathbb{P}^2(\mathbb{C})}_{\textcircled{2}} \cup \underbrace{\mathbb{P}^2(\mathbb{C})}_{\textcircled{3}} \cup \underbrace{\mathbb{P}^2(\mathbb{C})}_{\textcircled{4}}$$

$$\cup \underbrace{\mathbb{P}^1(\mathbb{C}) \times \mathbb{P}^2(\mathbb{C})}_{\textcircled{5}} \cup \underbrace{\{pt\}}_{\textcircled{6}}.$$

In general, $\dim V = n$.

$$\{ W \subseteq V \mid \dim W = m \}.$$

$\updownarrow$ $|\cdot|$

some geometric object,
(called Grassmannian).